

Week 9 Model Explainer

Equipment Failure Prediction

How the Model Works & When to Trust It

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1. The Analogy: How Does This Model Think?

Imagine you are an experienced senior mechanic who has serviced hundreds of machines over 20 years. When a machine comes in, you do not read one gauge and declare it healthy or failing. You check multiple things at once -- the vibration feel, the heat coming off the motor, the oil pressure, how long since it was last serviced, and how old the machine is. After years of experience, you have built an internal pattern: "When I see vibration this high AND the temperature is running hot AND it has not been serviced in 8 months -- that machine is going to fail soon."

The XGBoost model is that experienced mechanic -- but instead of intuition built over 20 years, it learned from 2,400 historical machine readings. It checks all 9 sensor inputs simultaneously and assigns a failure probability score between 0% and 100%.

A score above ~50% should be treated as a maintenance alert. A score above 80% means the machine needs immediate attention.

2. The Drivers: Top 3 Features Explained

The Feature Importance chart (see `feature_importance.png` in the repository) shows which sensor readings drive the model's predictions most strongly.

1st Driver -- Vibration (mm/s): Most Important

Vibration is the single strongest predictor of imminent failure. When a machine's vibration reading exceeds approximately 3.5 mm/s, the model sharply increases the failure probability.

Plain English: A machine that is shaking more than usual is signalling something is wrong -- a bearing wearing out, a shaft going out of balance, or a mounting loosening. High vibration readings precede approximately 35% of all failures in the training data.

2nd Driver -- Temperature (degrees C): Second Strongest Signal

Operating temperature above 90 degrees is the second strongest signal. Normal operating range is 65-85 degrees. Sustained high temperatures indicate inadequate cooling, friction buildup, or a blocked vent.

Plain English: A machine running hot is working too hard. Like a car engine overheating, sustained high temperature accelerates wear on seals, bearings, and lubricants.

3rd Driver -- Maintenance Lag (days since last service): Most Surprising

Machines that have not been serviced for more than 300 days show significantly elevated failure risk -- even when their current sensor readings look acceptable.

Plain English: A machine that has not been inspected in almost a year may have slowly degrading components that sensors do not fully capture. Think of it like skipping your car service: it may still drive fine, but the risk is accumulating quietly. The model learned that maintenance history is nearly as predictive as real-time sensor data -- operational discipline matters as much as the sensors themselves.

3. Trust & Limits: How the Operations Team Should Use This Tool

What the model does well

- Catches machines with multiple simultaneous warning signs -- combining all 9 sensors in a way no single threshold alarm can.
- Provides a ranked risk list so maintenance teams can prioritise which machines to check first when resources are limited.
- Explains WHY a machine was flagged via SHAP plots -- not just a black-box score.

Known Limitations

False Positives (False Alarms):

The model will occasionally flag machines that turn out to be fine. A high-vibration reading could come from a temporary external source (nearby construction, a passing forklift). Before raising a work order, the technician should physically inspect the machine and cross-check with recent maintenance logs.

False Negatives (Missed Failures):

Some failures happen suddenly without any gradual sensor drift -- a sudden bearing crack or power surge. The model cannot predict catastrophic instant failures. It is designed for gradual degradation patterns that develop over days to weeks.

Sensor Data Quality:

The model is only as reliable as the sensor readings fed into it. A faulty vibration sensor reporting 0.0 mm/s will make a failing machine appear healthy. Regular sensor calibration is essential for the model to remain accurate over time.

The Golden Rule

Use this model as a guide, not a replacement for engineering judgment. Treat a high-risk score as a strong recommendation to prioritise a physical inspection -- not a guaranteed failure prediction. A technician who disagrees with the model based on direct observation should always trust their hands-on judgment. The model augments human expertise -- it does not replace it.